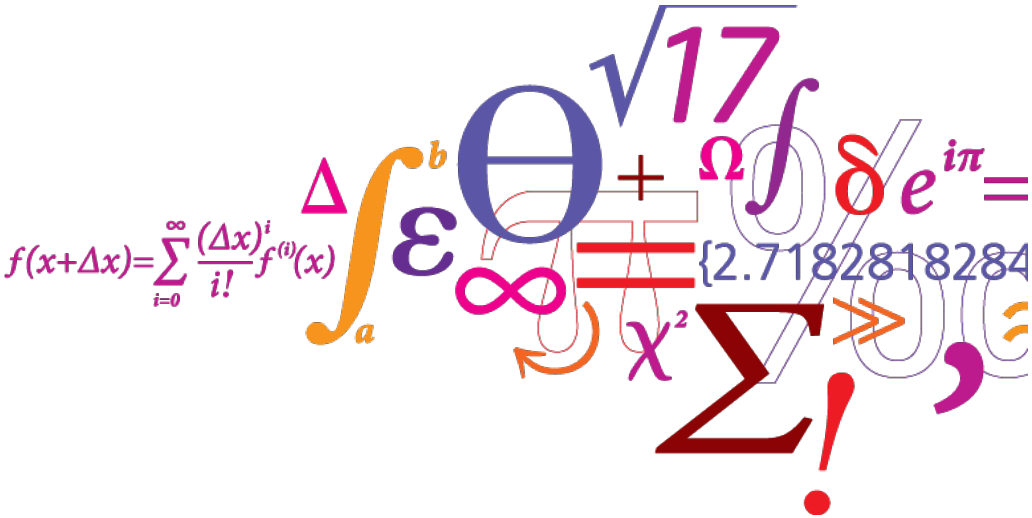
46755 – Renewables in Electricity Markets

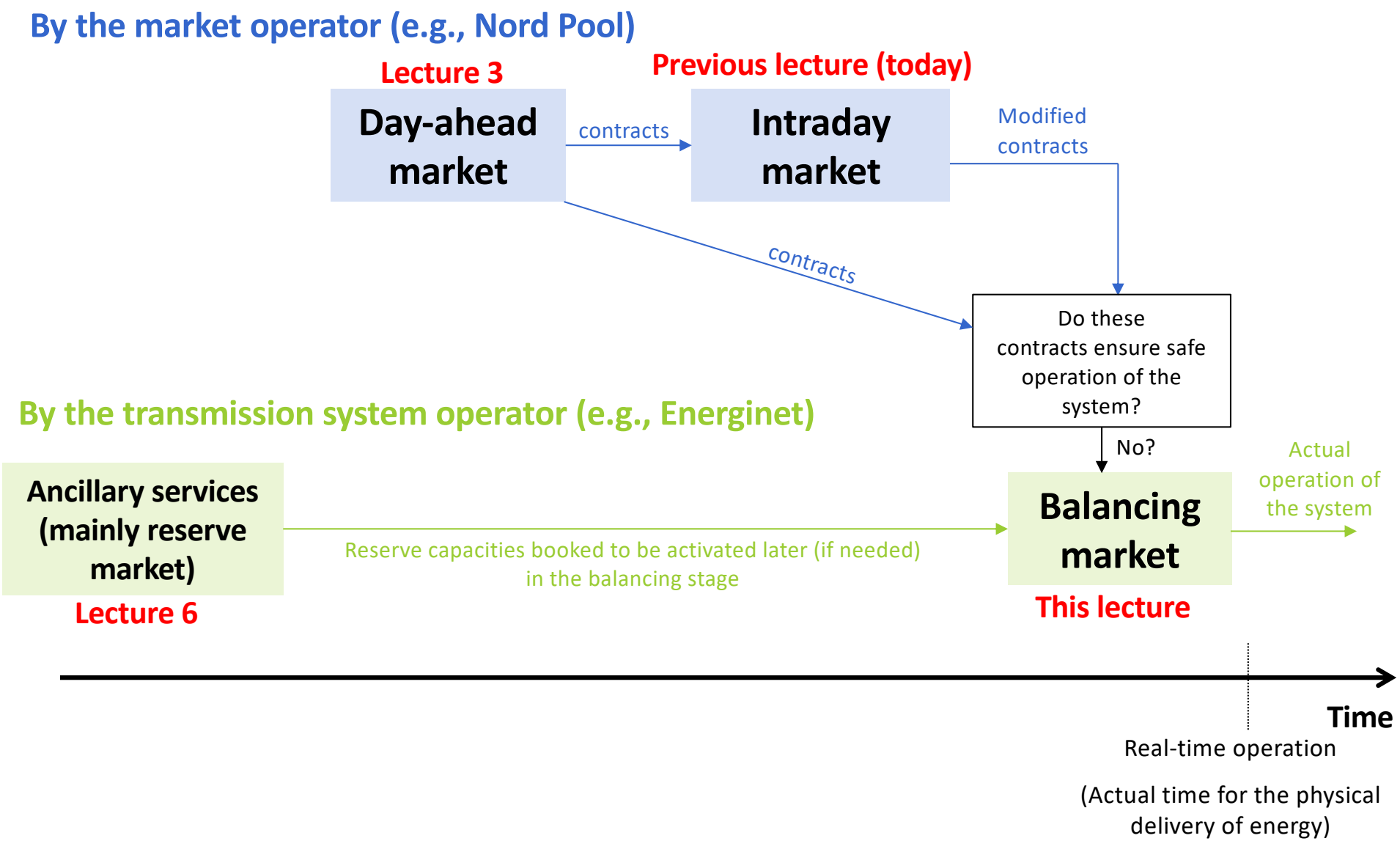
Lecture 5b: Balancing markets

Jalal Kazempour

March 3, 2025



Recap


By the market operator (e.g., Nord Pool)
Lecture 3
Previous lecture (today)
Day-ahead market
contracts
Intraday market
Modified contracts
contracts
Do these contracts ensure safe operation of the system?
No?
By the transmission system operator (e.g., Energinet)
Ancillary services (mainly reserve market)
Lecture 6
Reserve capacities booked to be activated later (if needed) in the balancing stage
Balancing market
This lecture
Actual operation of the system
Time
Real-time operation
(Actual time for the physical delivery of energy)

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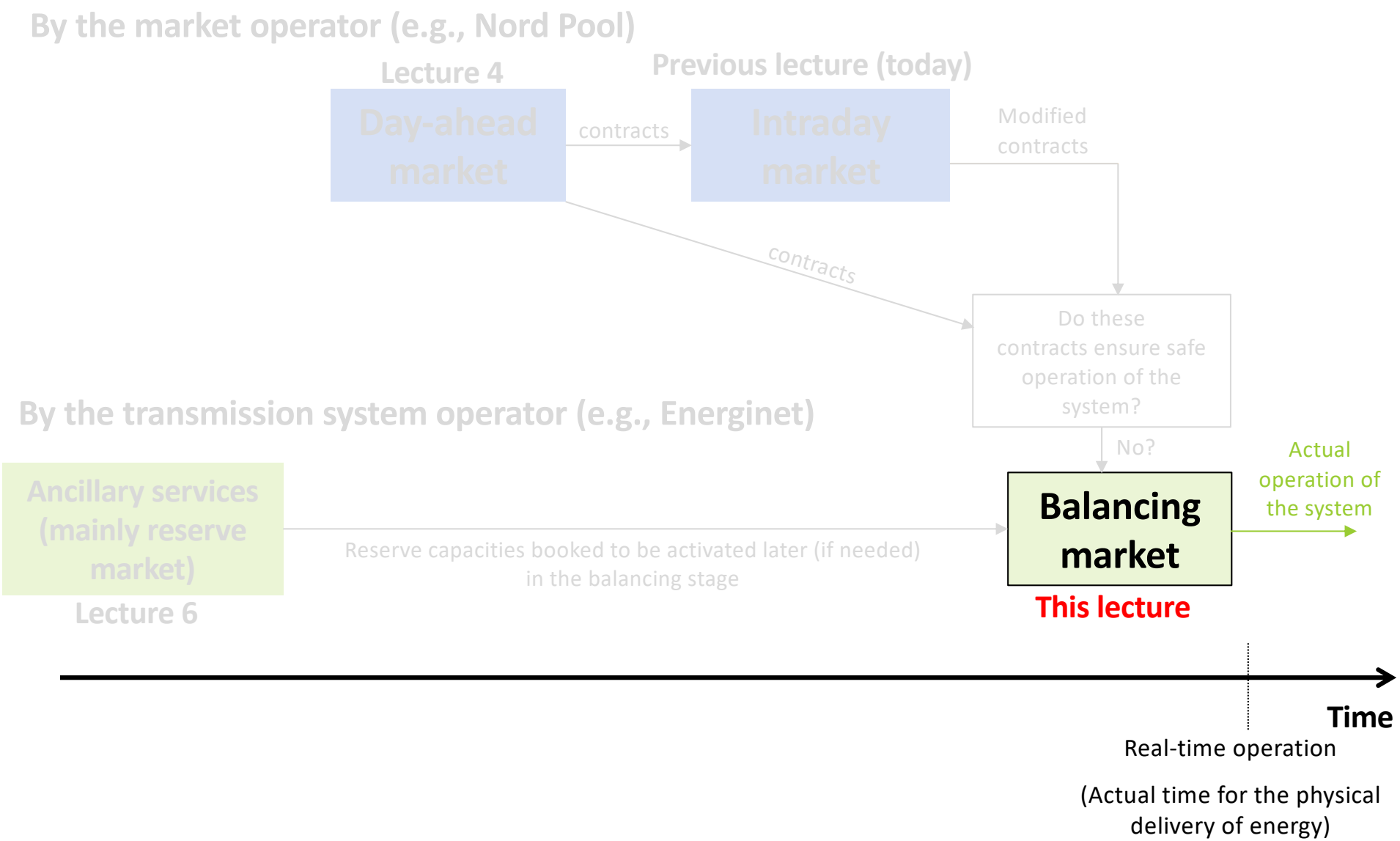
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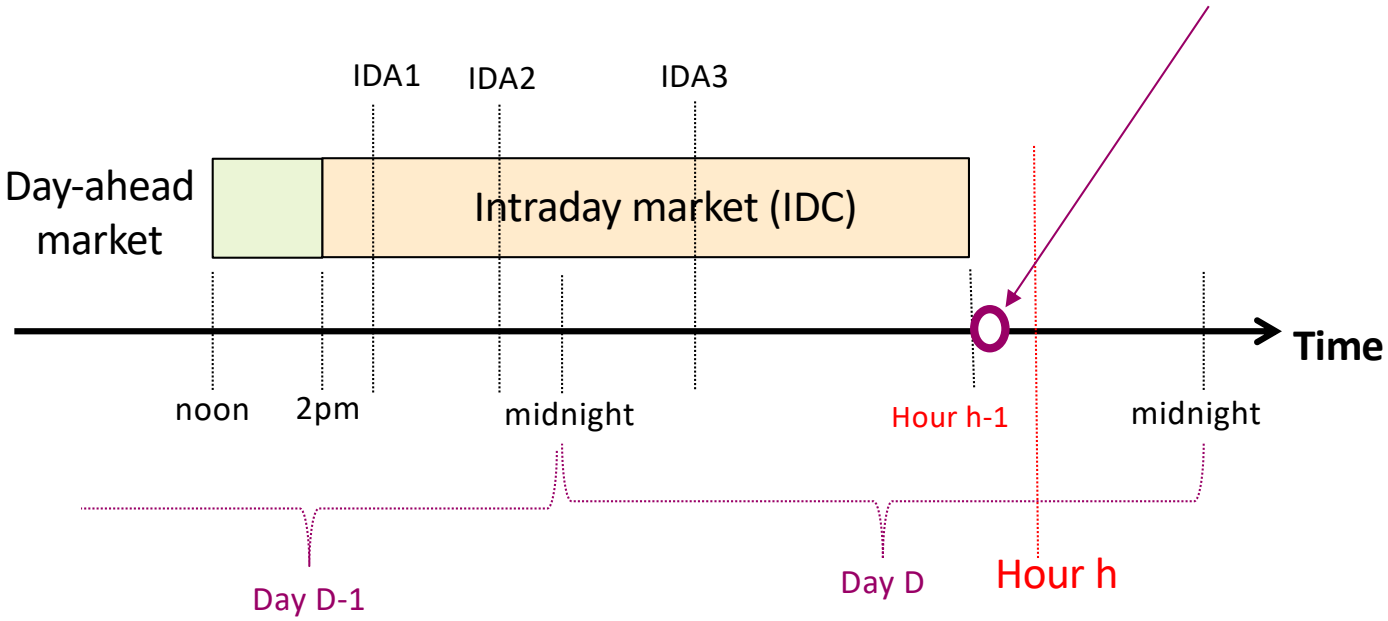
Learning objectives of this lecture

After this lecture, you are expected to:

- Explain the fundamentals of balancing markets.
- Formulate and solve the optimization problem for balancing market clearing.
- Calculate payments under both one-price and two-price balancing settlements.

Timeline

Consider a point in time when both the day-ahead and intraday markets have been cleared, and the power system must be operated for the upcoming hour (hour h) in less than one hour.



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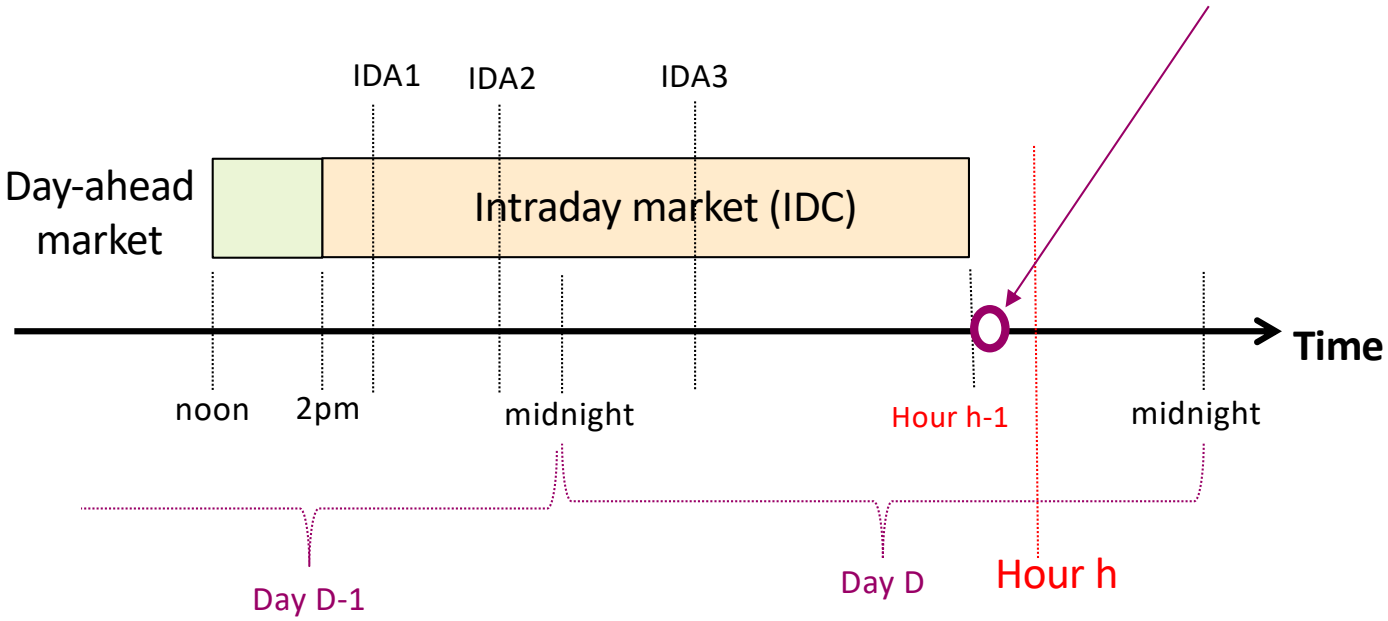

At this point, the market operator's role is complete, and it is now the **Transmission System Operator (TSO)** such as Energinet in Denmark who is responsible for operating the power system.

What happens at this point of time

There may still be power supply-demand imbalances in the underlying system due to:

- Errors in the forecast of renewable power generation and loads
- Unexpected failures of generation (or transmission) assets
- Planned imbalances if they are envisioned to be profitable by certain players (there might be investigations if the imbalance is significant)
- etc.

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Balancing Market
to be cleared by the TSO

From the TSO perspective

There are two types of agents in the balancing market:

From the TSO perspective

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1) **Balancing service providers (BSPs):**
   - Any generation- or demand-side market participant capable of adjusting their set-point in real time to help balance the system.
   - Without a properly structured reserve market in advance to secure enough capacity, there may be a scarcity of balancing services during real-time operation (which will be discussed in the next lecture).

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2) **Balancing responsible parties (BRPs):**
   - In European electricity markets, BRPs have agreements with system operators and bear "financial" responsibilities. These BRPs can even be financial institutions or banks.
   - Each producer or demand-side participant must assign a BRP, who assumes financial responsibility for any mismatch between the scheduled and actual production/consumption in real time.

From the TSO perspective

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BSPs are paid for balancing service provision.

BRPs are charged for imbalances caused.

Energinet's definition of BRPs

BALANCE-RESPONSIBLE PARTIES

Balance-responsible parties, or BRPs, buy and sell electricity on power exchanges on behalf of electricity suppliers and facility owners. Every day, BRPs submit plans to Energinet on the volume of electricity expected to be generated and consumed in the next 24 hours by the producers and customers that the respective BRP is balance-responsible party for.

BRPs are financially accountable to Energinet for imbalances between expected and actual generation and consumption on the day of operation. Actually, it is impossible to predict a day's generation and consumption a hundred per cent accurately. Any imbalance will be balanced out by Energinet, and BRPs are then invoiced the price of the respective imbalances that they caused.

Source: Energinet (https://en.energinet.dk/Electricity/New-player/Roles-and-responsibilities)

Power imbalances

In the case of a "power deficit" in the system (caused, for example, by a generator failure or under-production of renewables)

Upward regulation is needed:

- increased production by generation-side BSPs
- reduced consumption by demand-side BSPs

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Upward regulation is needed:

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In the case of a "power excess" in the system (caused, for example, by over-production of renewables)

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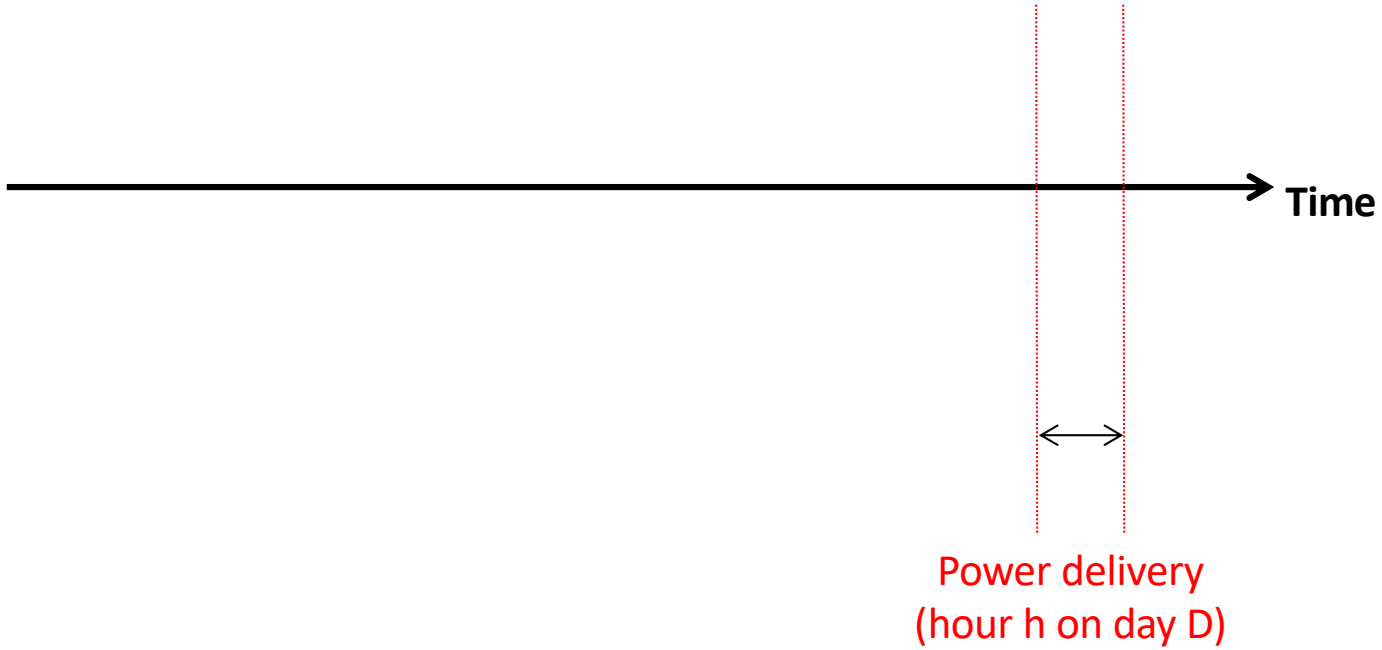
Downward regulation is needed:

- reduced production by generation-side BSPs
- increased consumption by demand-side BSPs

A full overview of imbalance management

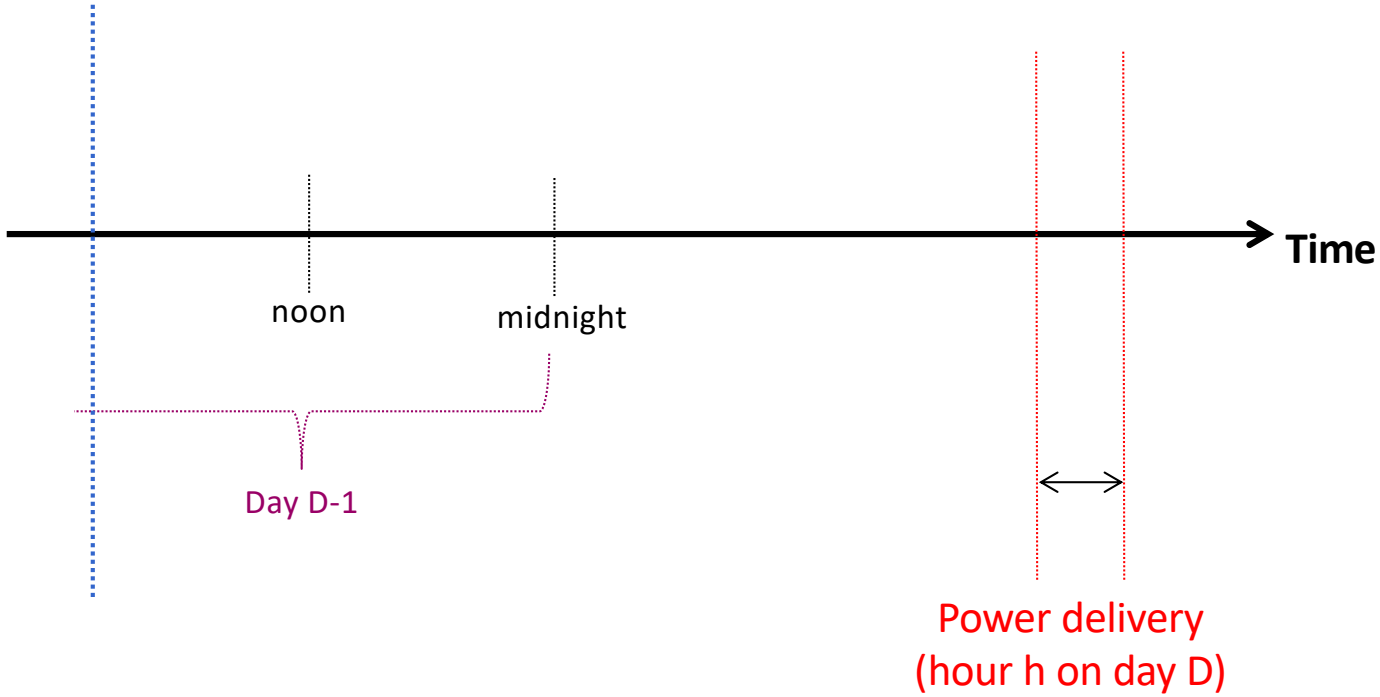


A full overview of imbalance management

Reserve Markets

The TSO procures various ancillary services from BSPs for each hour h on day D, including Frequency Restoration Reserve (FRR) services.

(Details will be provided in the next lecture)



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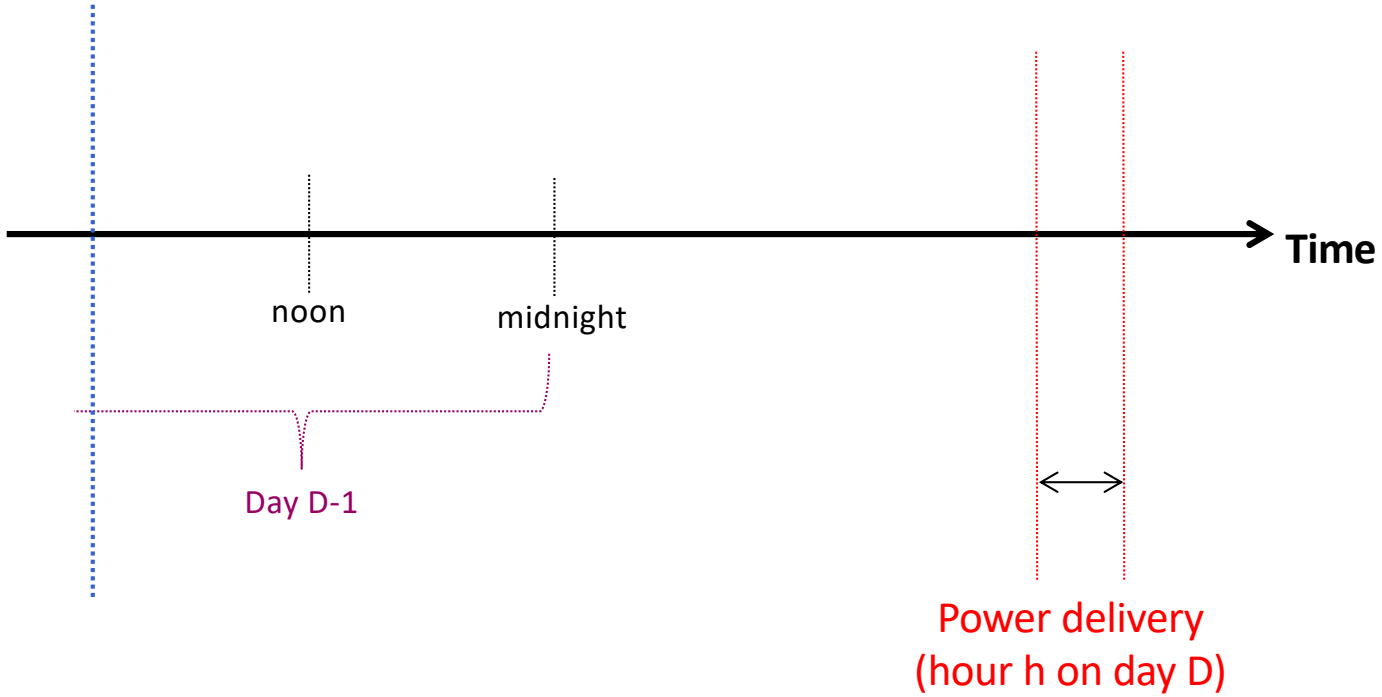
Based on reserve market outcomes, some BSPs are allocated for FRR services.

- Example: A gas-fired power plant may have allocated 1 MW h for upward regulation during hour h on day D.



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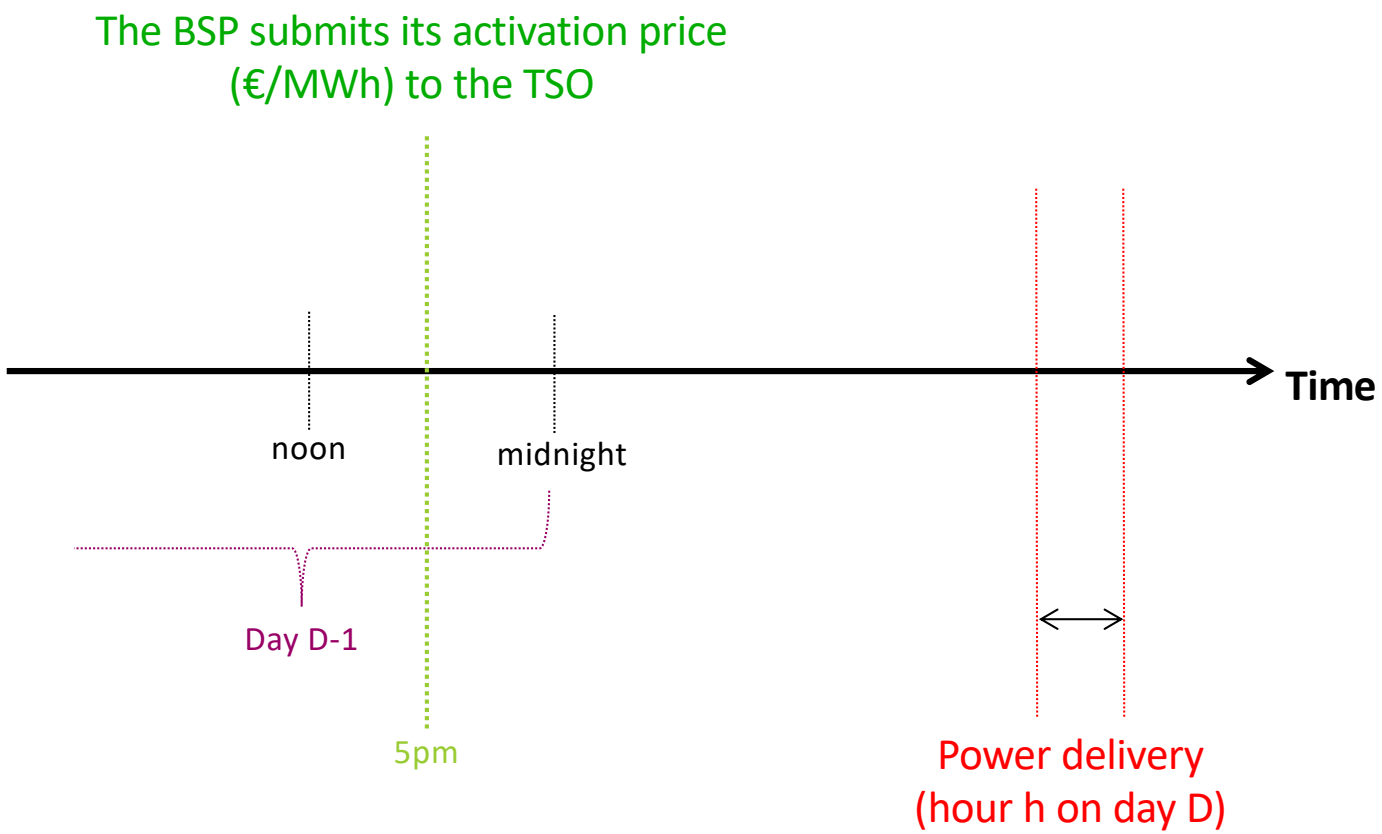

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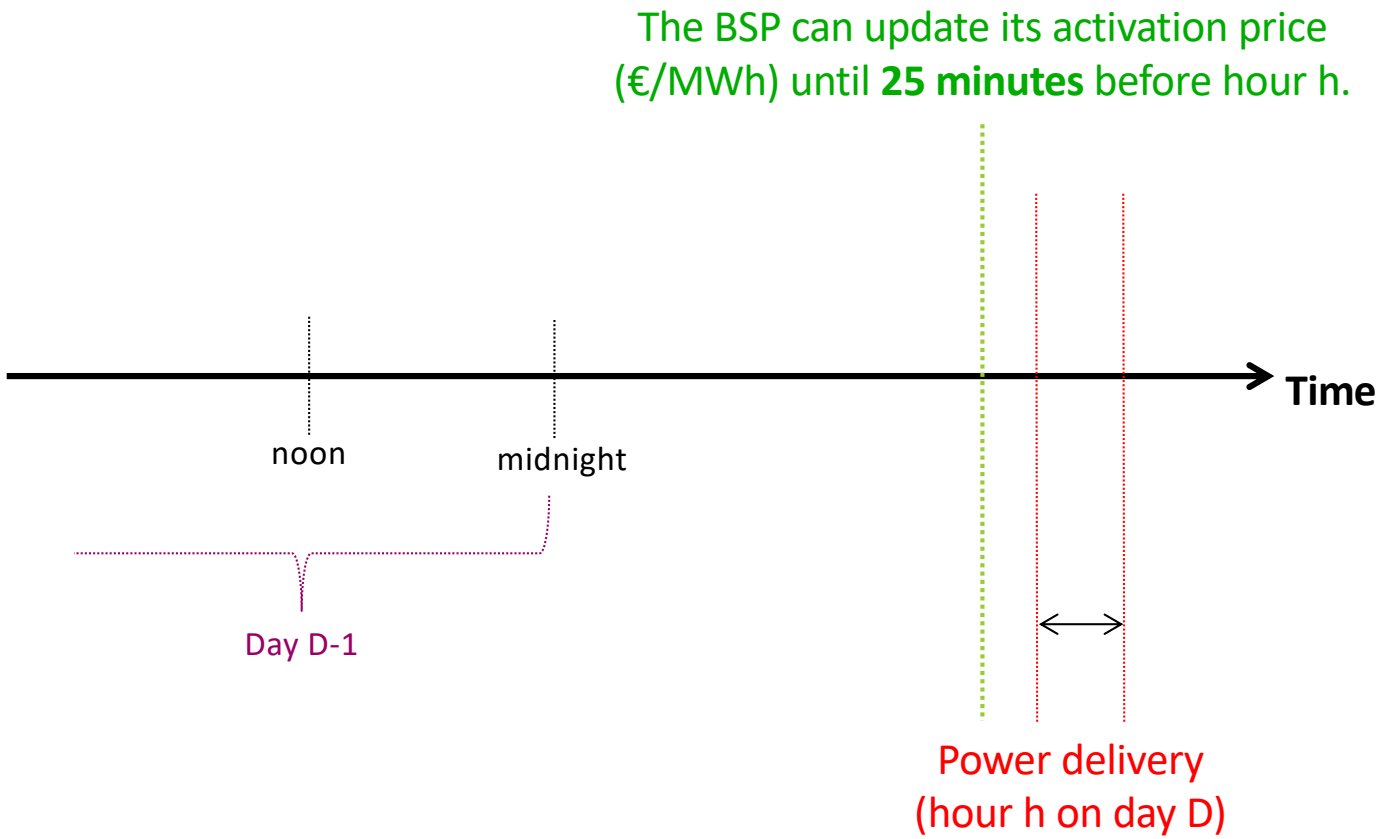


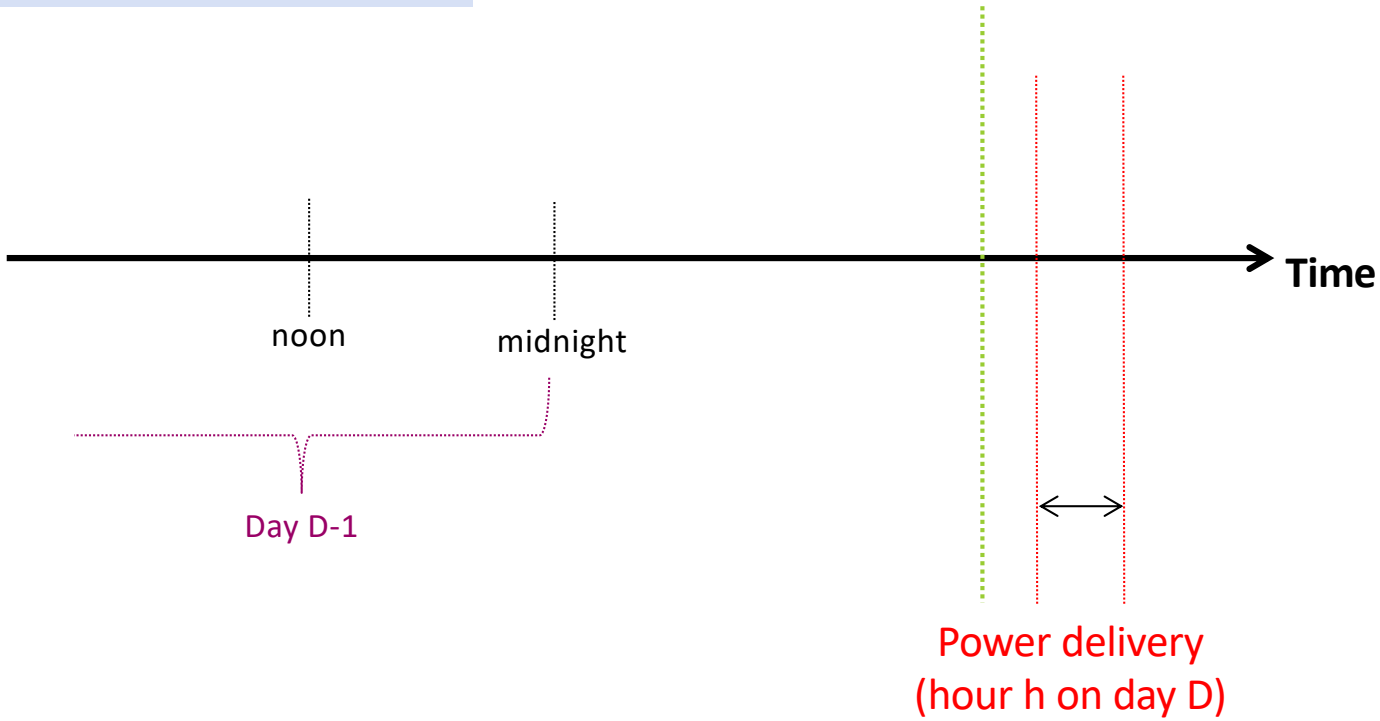
A full overview of imbalance management

Voluntary bids

(from BSPs who have **not** already been reserved in the reserve market)

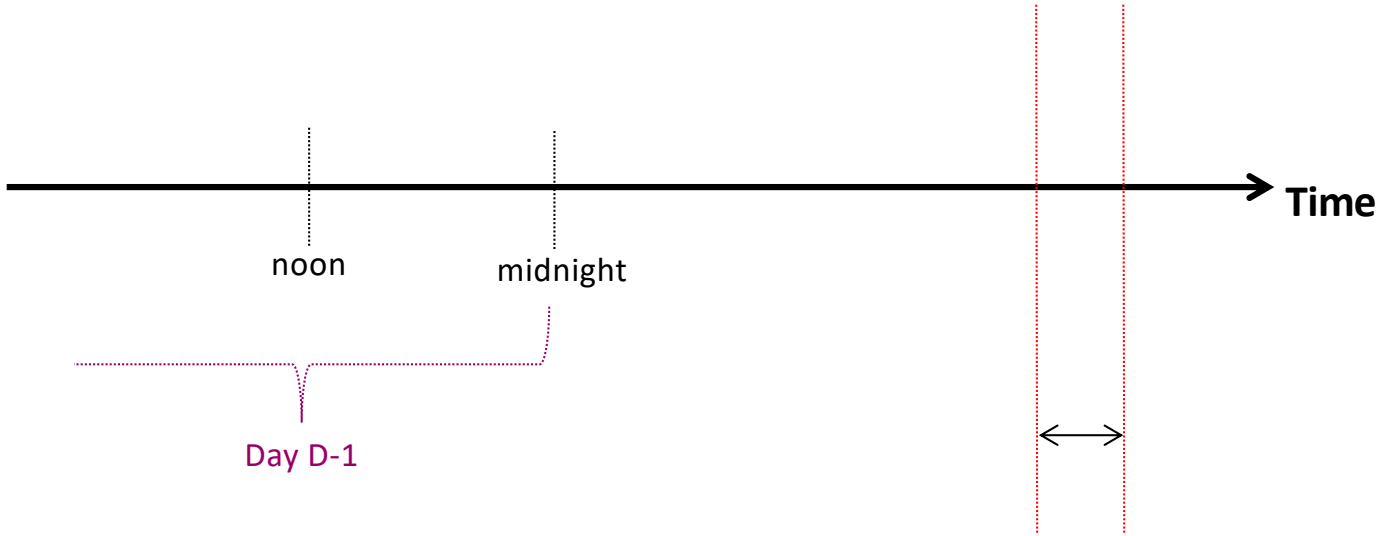
can also be submitted, including the reserve quantity and activation price.

The BSP can update its activation price (€/MWh) until **25 minutes** before hour h.



A full overview of imbalance management

During hour h, the TSO activates the BSPs, if needed, based on their activation price, following merit order principles.



Power delivery
(hour h on day D)

A full overview of imbalance management

Balancing price

The balancing price is set to the activation price of the most expensive BSP activated!

Note: Market time unit (granularity) will be 15 minutes starting from March 2025.

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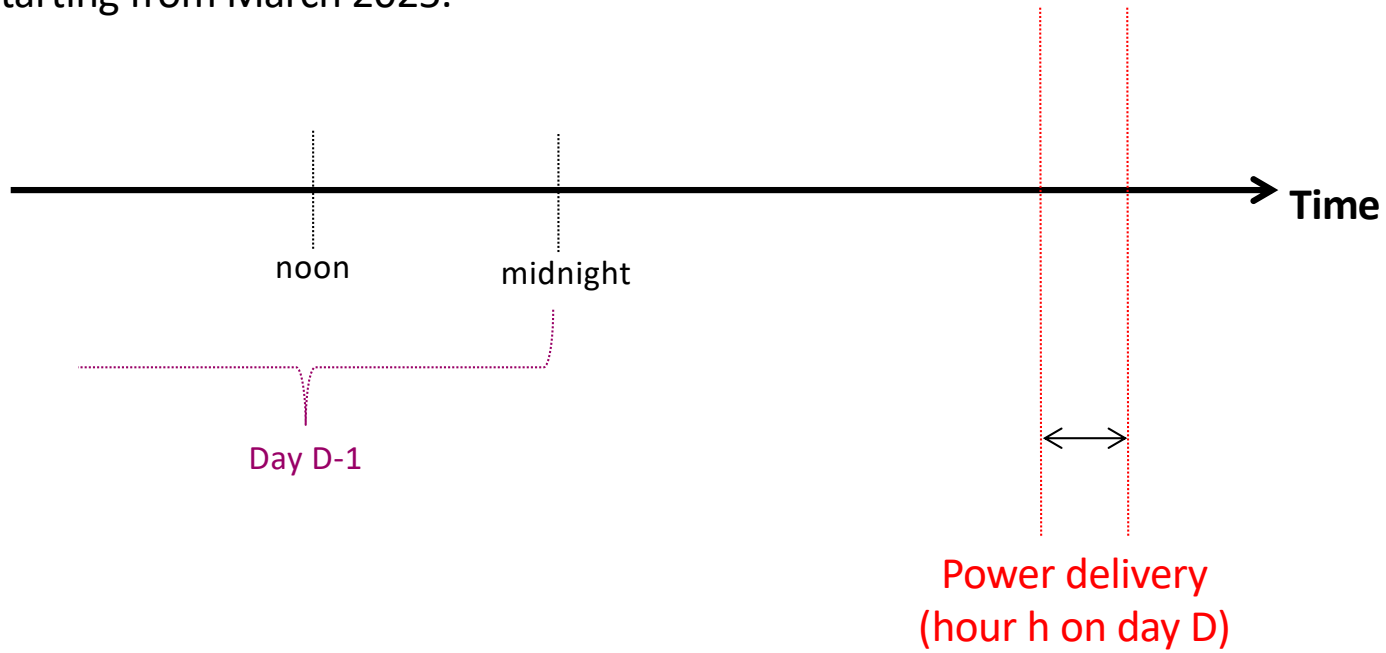

A full overview of imbalance management

Balancing price

The balancing price is set to the activation price of the most expensive BSP activated!

Note: Market time unit (granularity) will be 15 minutes starting from March 2025.

The balancing price setting will be revised to account for the activation of both manual (mFRR) and automatic (aFRR) services.

- Not to be discussed in this course.
- Further info: https://energinet.dk/media/i5bd2yg4/questions-and-answers-to-nordic-imbalance-pricing-webinar.pdf

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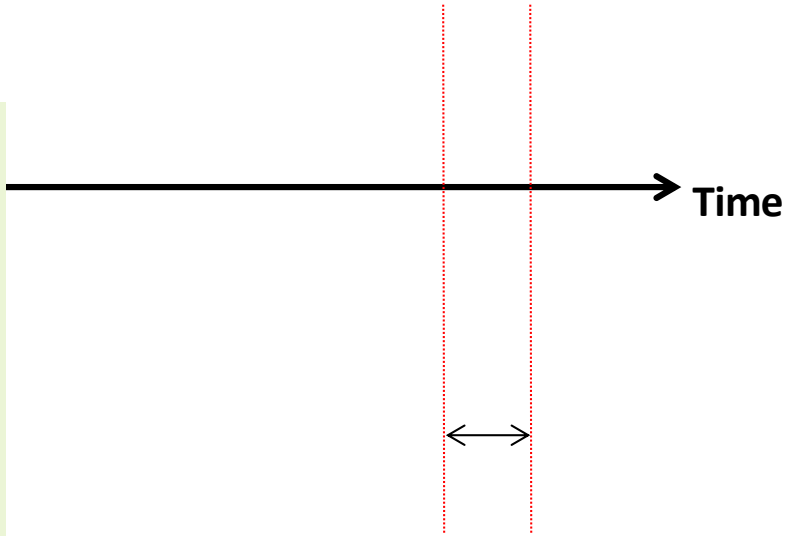

Power delivery
(hour h on day D)

An illustrative example

Day-ahead forecast: 20 MW
Offer price: €0/MWh

Capacity: 50 MW
Offer price: €20/MWh

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

For the sake of simplicity, let's discard the intraday market.

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Example 1: Due to an unexpected outage, G1 is out of the system during real-time operation.

→ System imbalance = +50 MW (upward regulation is needed)

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Example 2: The wind power producer, W1, generates 35 MW during real-time operation.
→ System imbalance = -15 MW (downward regulation is needed).

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Example 2: The wind power producer, W1, generates 35 MW during real-time operation.
→ System imbalance = -15 MW (downward regulation is needed).

Example 3: A combination of Examples 1 and 2

→ System imbalance = +35 MW (upward regulation is needed).

- Although W1 causes an imbalance, it results in a "positive imbalance". This means it produces "unplanned" extra power when the system needs it. Here, a positive imbalance implies a **desired** imbalance.

An illustrative example

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Capacity: 100 MW
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Demand D1

Maximum load: 90 MW
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- In the worst case, if there is insufficient upward balancing service, demand D1 can be involuntarily curtailed as an extreme balancing action. However, the corresponding cost, known as the "curtailment cost" or "load shedding cost," is generally extremely high.

- On the other hand, if there is insufficient downward balancing service, the excess wind power can be spilled, usually at zero cost.

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Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

Remember, the day-ahead market-clearing price is €30/MWh.

Question: Assuming G2 and D1 are "rational" market participants, will they submit their activation price for downward regulation at a price higher than or lower than the day-ahead market price (€30/MWh)?

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Answer: Lower!

- Generator G2 is already paid for her production schedule (20 MW) in the day-ahead market at €30/MWh. By providing downward regulation, it will produce less than scheduled (e.g., 15 MW), and therefore it must pay back in the balancing market for the power not produced (5 MW). As a result, it will submit the activation price for downward regulation at a price lower than €30/MWh to earn money.
- A similar reasoning applies to demand D1.

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Answer: Higher (or at least equal)!

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Question: What are the maximum quantities of upward and downward regulation (in MW) that generator G2 can offer?

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Answer: Its day-ahead schedule is 20 MW

→ Maximum quantity of downward regulation = 20 MW

→ Maximum quantity of upward regulation = 100 – 20 = 80 MW

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→ Maximum quantity of downward regulation = 20 MW

→ Maximum quantity of upward regulation = 100 – 20 = 80 MW

A generator can provide downward balancing service only if it is already scheduled in the day-ahead market. A generator with a day-ahead schedule of 0 MW cannot provide downward regulation in the balancing stage. This is a critical issue that the system operator must be aware of!

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Answer: Its day-ahead schedule is 90 MW

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Answer: Its day-ahead schedule is 90 MW

→ Maximum quantity of downward regulation = 0 MW

→ Maximum quantity of upward regulation = 90 MW

A demand can provide downward balancing service only if it is not *fully* supplied in the day-ahead market.

An illustrative example

W1

Day-ahead forecast: 20 MW
Offer price: €0/MWh

G1 POWER PLANT

Capacity: 50 MW
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Maximum load: 90 MW
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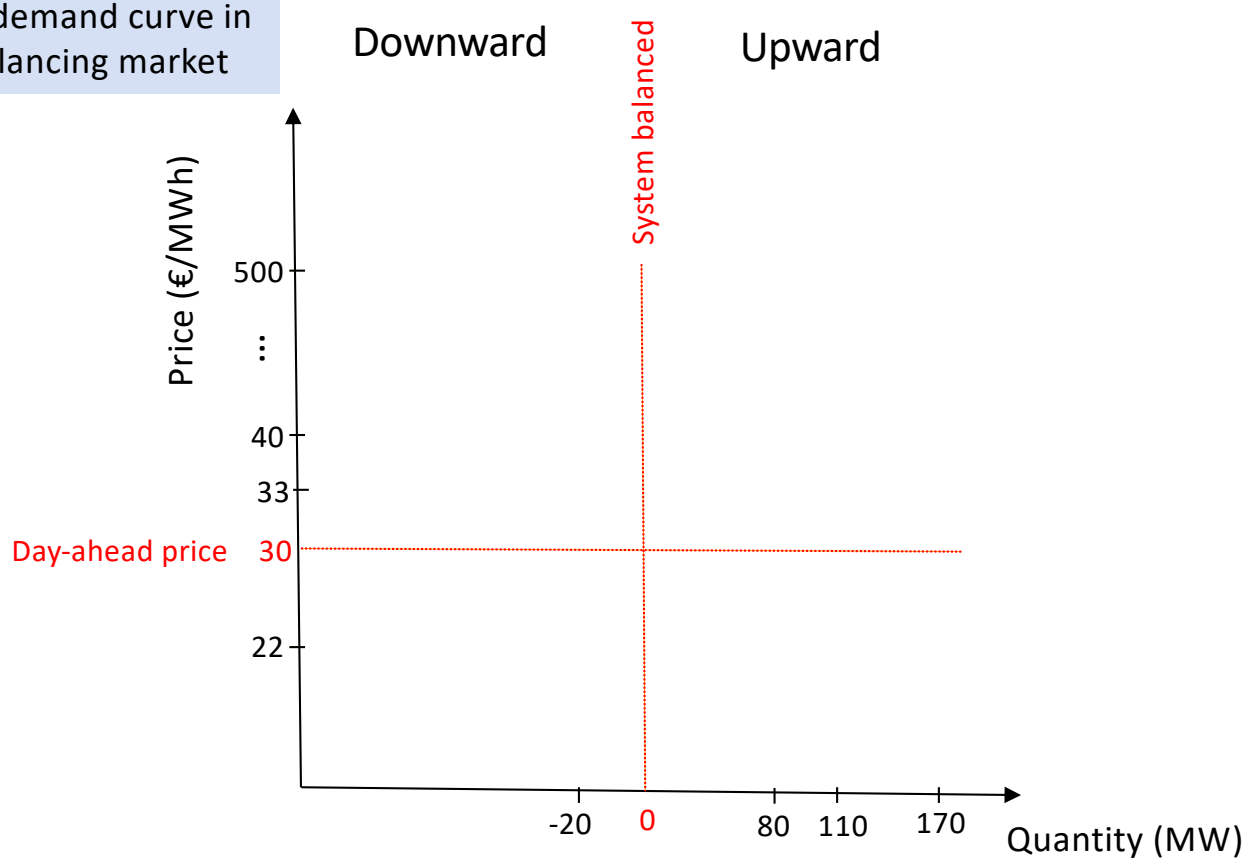
Supply-demand curve in the balancing market



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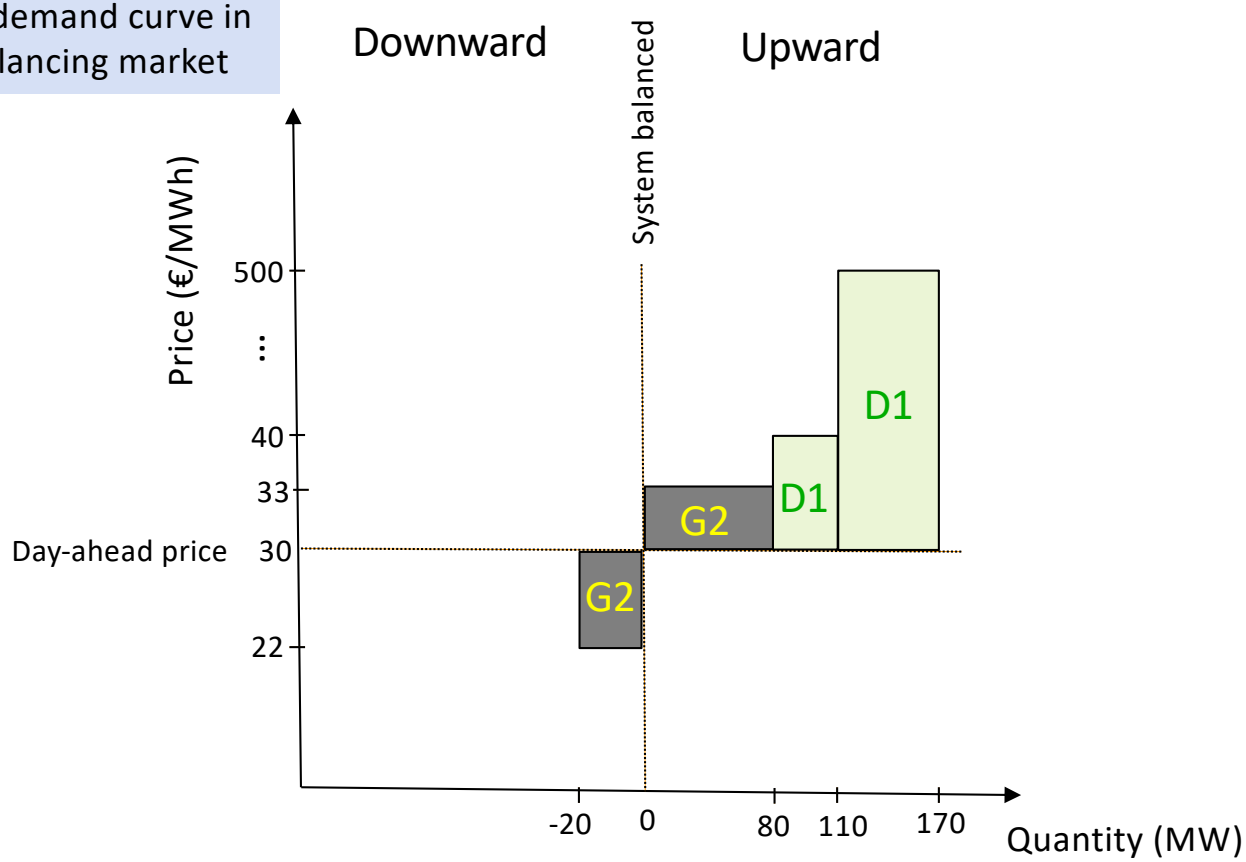

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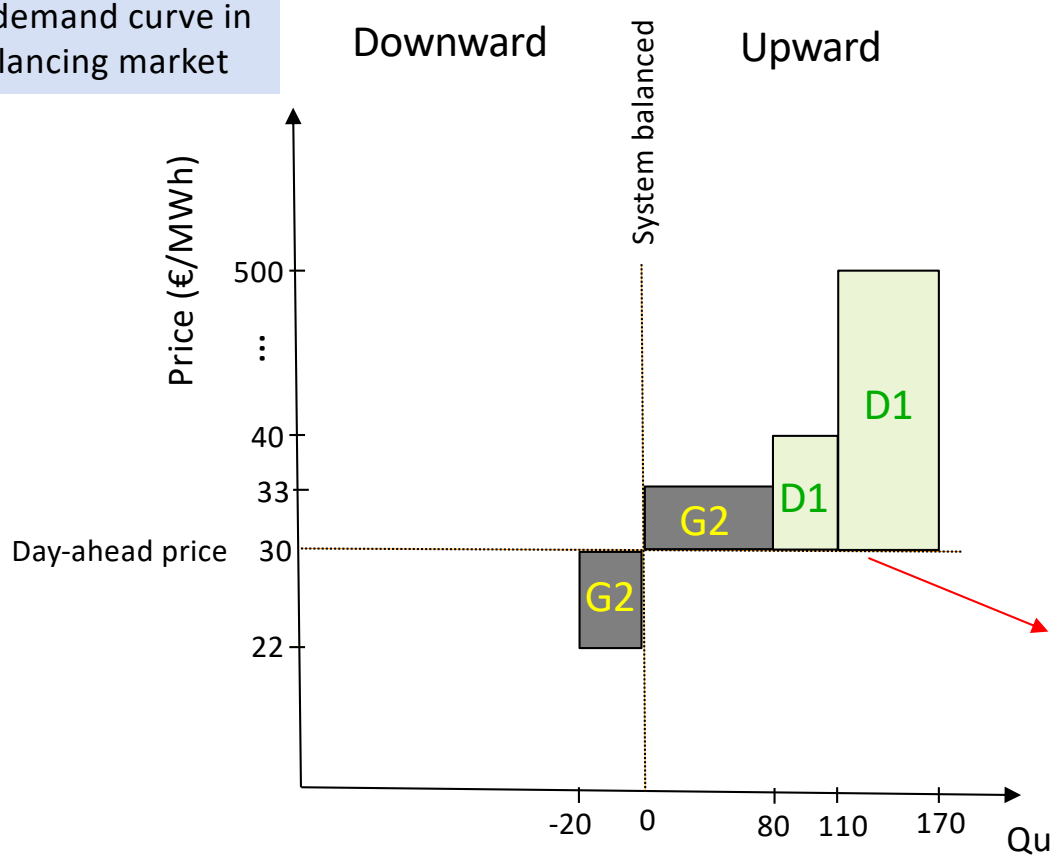

Let's assume D1 offers an upward regulation service of a maximum of 30 MW at a price of €40/MWh. The remaining 60 MW of demand, if curtailed, should be paid at a very high price (€500/MWh)!

An illustrative example

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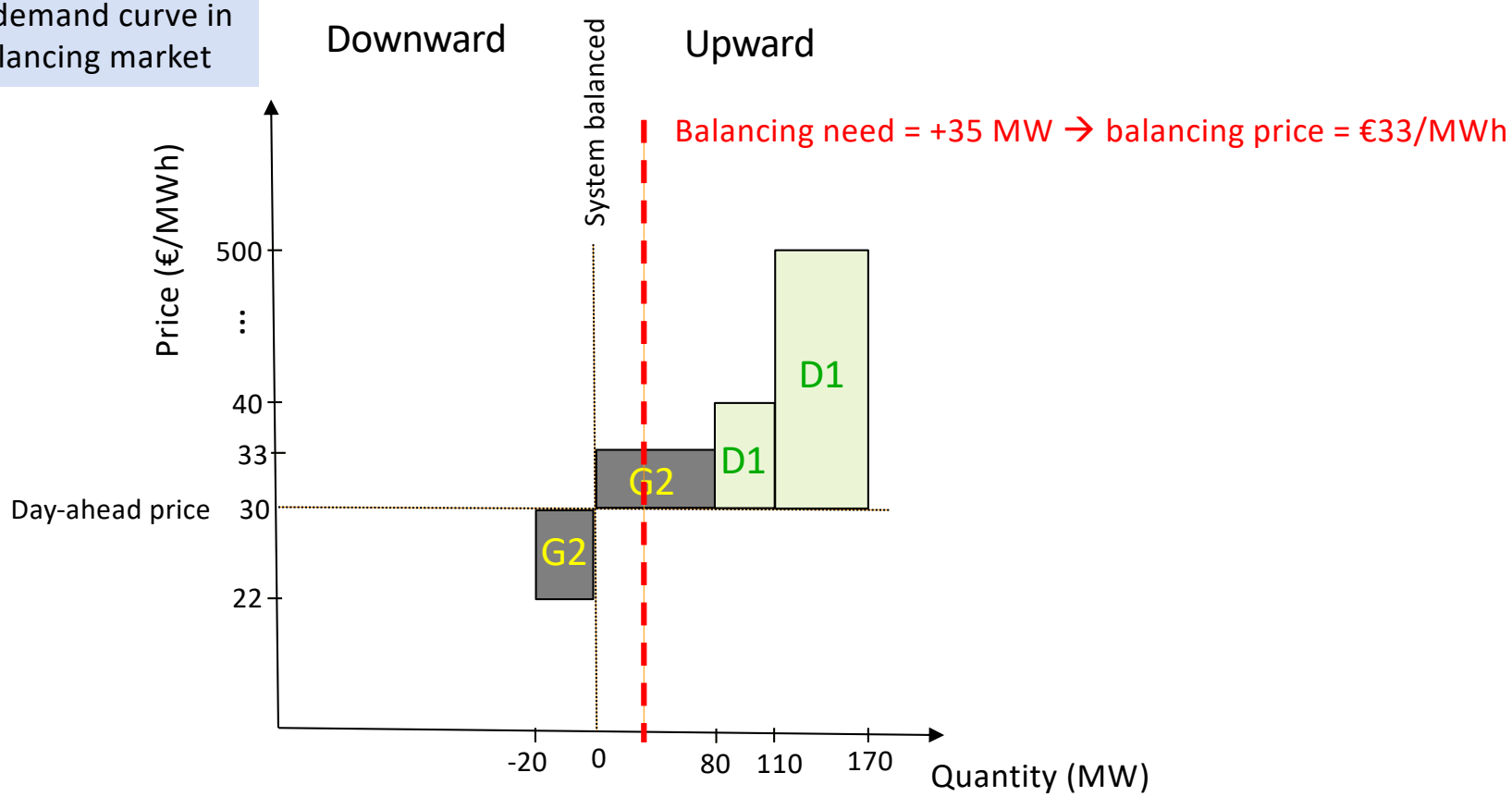

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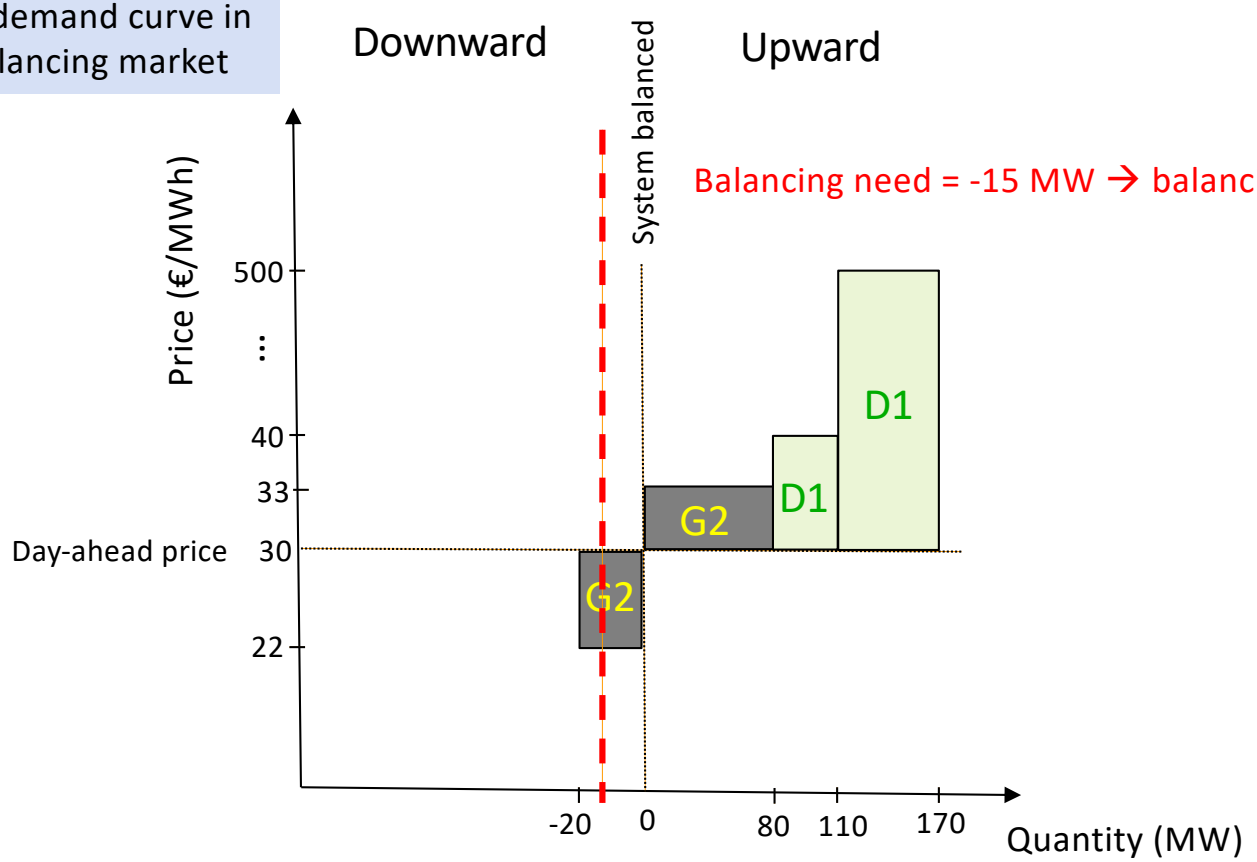

Balancing need = -15 MW → balancing price = €22/MWh

An illustrative example

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Maximum load: 90 MW
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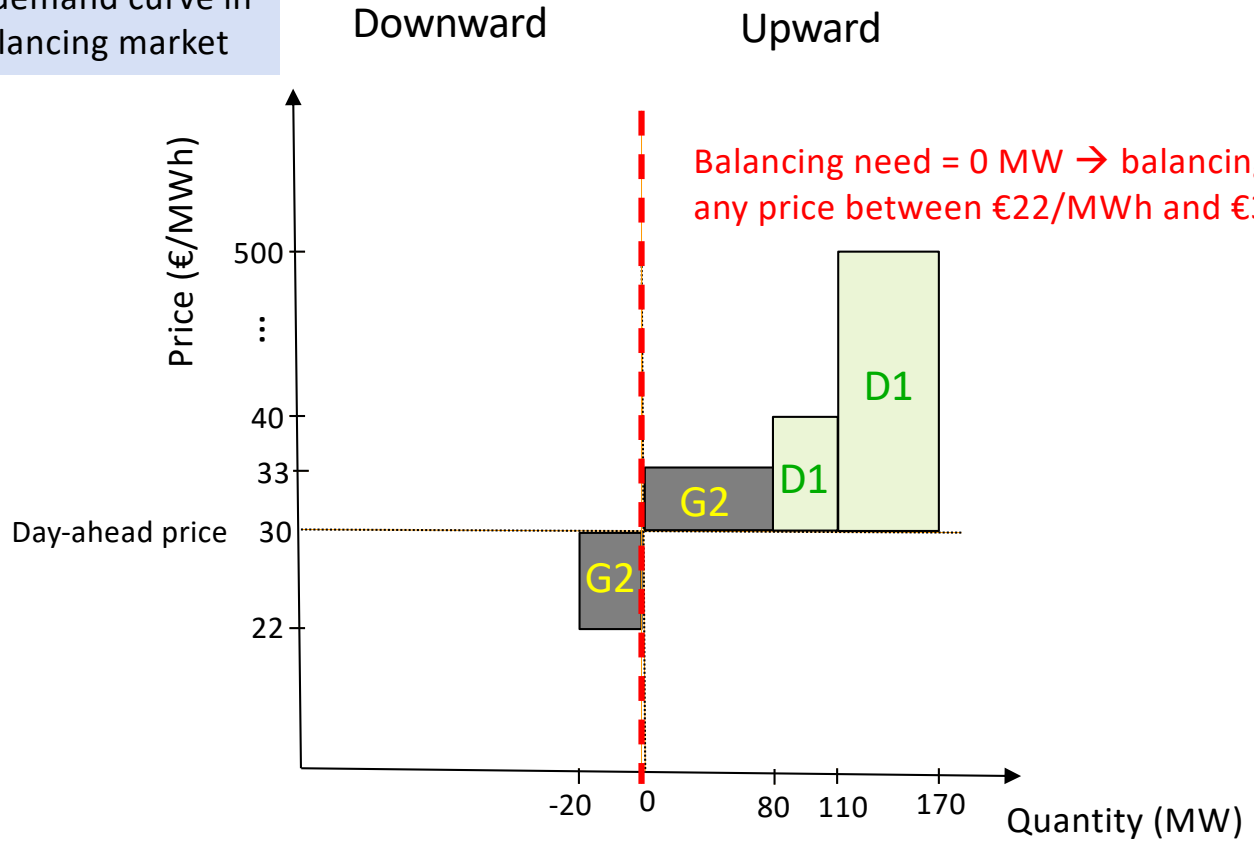
Supply-demand curve in the balancing market



Balancing need = 0 MW → balancing price = €30/MWh (accurately, any price between €22/MWh and €33/MWh)!

An illustrative example

Day-ahead forecast: 20 MW
Offer price: €0/MWh

Capacity: 50 MW
Offer price: €20/MWh

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

Takeaways:

- ✓ Balancing need > 0 → Upward regulation is needed → Balancing price > Day-ahead price
- ✓ Balancing need < 0 → Downward regulation is needed → Balancing price < Day-ahead price

Optimization for balancing market clearing

$$\underset{\boxed{p^{\mathrm{G2}\downarrow}},\ p^{\mathrm{G2}\uparrow},\ p^{\mathrm{D1}\uparrow},\ p^{\mathrm{D1,curt}}}{\text{Minimize}} \quad 33\ p^{\mathrm{G2}\uparrow} + 40\ p^{\mathrm{D1}\uparrow} + 500\ p^{\mathrm{D1,curt}} - 22\ p^{\mathrm{G2}\downarrow}$$

Downward regulation of generator G2

subject to:

$$p^{\mathrm{G2}\uparrow} + p^{\mathrm{D1}\uparrow} + p^{\mathrm{D1,curt}} - p^{\mathrm{G2}\downarrow} = 35 \quad : \quad \lambda^{\mathrm{Balancing}}$$

$$0 \leq p^{\mathrm{G2}\uparrow} \leq 80$$

$$0 \leq p^{\mathrm{D1}\uparrow} \leq 30$$

$$0 \leq p^{\mathrm{D1,curt}} \leq 60$$

$$0 \leq p^{\mathrm{G2}\downarrow} \leq 20$$

Optimization for balancing market clearing

$$\underset{p^{G2\downarrow},\ \boxed{p^{G2\uparrow}},\ p^{D1\uparrow},\ p^{D1,curt}}{\text{Minimize}} \quad 33\ p^{G2\uparrow} + 40\ p^{D1\uparrow} + 500\ p^{D1,curt} - 22\ p^{G2\downarrow}$$

Upward regulation of generator G2

subject to:

$$p^{G2\uparrow} + p^{D1\uparrow} + p^{D1,curt} - p^{G2\downarrow} = 35 \quad : \quad \lambda^{Balancing}$$

$$0 \leq p^{G2\uparrow} \leq 80$$

$$0 \leq p^{D1\uparrow} \leq 30$$

$$0 \leq p^{D1,curt} \leq 60$$

$$0 \leq p^{G2\downarrow} \leq 20$$

Optimization for balancing market clearing

$$\underset{p^{G2\downarrow},\ p^{G2\uparrow},\ \boxed{p^{D1\uparrow}},\ p^{D1,curt}}{\text{Minimize}} \quad 33\ p^{G2\uparrow} + 40\ p^{D1\uparrow} + 500\ p^{D1,curt} - 22\ p^{G2\downarrow}$$

Upward regulation of demand D1

subject to:

$$p^{G2\uparrow} + p^{D1\uparrow} + p^{D1,curt} - p^{G2\downarrow} = 35 \quad : \quad \lambda^{Balancing}$$

$$0 \leq p^{G2\uparrow} \leq 80$$

$$0 \leq p^{D1\uparrow} \leq 30$$

$$0 \leq p^{D1,curt} \leq 60$$

$$0 \leq p^{G2\downarrow} \leq 20$$

Optimization for balancing market clearing

$$\underset{p^{G2\downarrow},\ p^{G2\uparrow},\ p^{D1\uparrow},\ \boxed{p^{D1,curt}}}{\text{Minimize}} \quad 33\ p^{G2\uparrow} + 40\ p^{D1\uparrow} + 500\ p^{D1,curt} - 22\ p^{G2\downarrow}$$

Involuntarily load curtailment of demand D1

subject to:

$$p^{G2\uparrow} + p^{D1\uparrow} + p^{D1,curt} - p^{G2\downarrow} = 35 \quad : \quad \lambda^{Balancing}$$

$$0 \leq p^{G2\uparrow} \leq 80$$

$$0 \leq p^{D1\uparrow} \leq 30$$

$$0 \leq p^{D1,curt} \leq 60$$

$$0 \leq p^{G2\downarrow} \leq 20$$

Optimization for balancing market clearing

$$\underset{p^{\mathrm{G2}\downarrow},\ p^{\mathrm{G2}\uparrow},\ p^{\mathrm{D1}\uparrow},\ p^{\mathrm{D1,curt}}}{\text{Minimize}} \quad \underbrace{33\ p^{\mathrm{G2}\uparrow} + 40\ p^{\mathrm{D1}\uparrow} + 500\ p^{\mathrm{D1,curt}}} - 22\ p^{\mathrm{G2}\downarrow}$$

Total cost of upward regulation based on the activation price of BSPs (increased production, reduced demand)

subject to:

$$p^{\mathrm{G2}\uparrow} + p^{\mathrm{D1}\uparrow} + p^{\mathrm{D1,curt}} - p^{\mathrm{G2}\downarrow} = 35 \quad : \quad \lambda^{\mathrm{Balancing}}$$

$$0 \leq p^{\mathrm{G2}\uparrow} \leq 80$$

$$0 \leq p^{\mathrm{D1}\uparrow} \leq 30$$

$$0 \leq p^{\mathrm{D1,curt}} \leq 60$$

$$0 \leq p^{\mathrm{G2}\downarrow} \leq 20$$

Optimization for balancing market clearing

$$\underset{p^{G2\downarrow},\ p^{G2\uparrow},\ p^{D1\uparrow},\ p^{D1,curt}}{\text{Minimize}} \quad 33\ p^{G2\uparrow} + 40\ p^{D1\uparrow} + 500\ p^{D1,curt} - \underbrace{22\ p^{G2\downarrow}}$$

Corresponding to the downward regulation
Recall, G2 will pay back if activated for downward regulation

subject to:

$$p^{G2\uparrow} + p^{D1\uparrow} + p^{D1,curt} - p^{G2\downarrow} = 35 \quad : \quad \lambda^{Balancing}$$

$$0 \leq p^{G2\uparrow} \leq 80$$

$$0 \leq p^{D1\uparrow} \leq 30$$

$$0 \leq p^{D1,curt} \leq 60$$

$$0 \leq p^{G2\downarrow} \leq 20$$

Optimization for balancing market clearing

$$\underset{p^{G2\downarrow},\ p^{G2\uparrow},\ p^{D1\uparrow},\ p^{D1,curt}}{\text{Minimize}} \quad 33\ p^{G2\uparrow} + 40\ p^{D1\uparrow} + 500\ p^{D1,curt} - 22\ p^{G2\downarrow}$$

subject to:

$$p^{G2\uparrow} + p^{D1\uparrow} + p^{D1,curt} - p^{G2\downarrow} = 35 \quad : \quad \lambda^{Balancing}$$

Total balancing services provided = balancing need (here, 35 MW)

$$0 \leq p^{G2\uparrow} \leq 80$$

$$0 \leq p^{D1\uparrow} \leq 30$$

$$0 \leq p^{D1,curt} \leq 60$$

$$0 \leq p^{G2\downarrow} \leq 20$$

Limits of balancing service providers

Optimization for balancing market clearing

$$\underset{p^{\mathrm{G2}\downarrow},\ p^{\mathrm{G2}\uparrow},\ p^{\mathrm{D1}\uparrow},\ p^{\mathrm{D1,curt}}}{\text{Minimize}} \quad 33\ p^{\mathrm{G2}\uparrow} + 40\ p^{\mathrm{D1}\uparrow} + 500\ p^{\mathrm{D1,curt}} - 22\ p^{\mathrm{G2}\downarrow}$$

subject to:

$$p^{\mathrm{G2}\uparrow} + p^{\mathrm{D1}\uparrow} + p^{\mathrm{D1,curt}} - p^{\mathrm{G2}\downarrow} = 35 \quad : \quad \boxed{\lambda^{\mathrm{Balancing}}}$$

Balancing price (dual variable)

$$0 \leq p^{\mathrm{G2}\uparrow} \leq 80$$

$$0 \leq p^{\mathrm{D1}\uparrow} \leq 30$$

$$0 \leq p^{\mathrm{D1,curt}} \leq 60$$

$$0 \leq p^{\mathrm{G2}\downarrow} \leq 20$$

Optimization for balancing market clearing

In a **general** and **compac**t form (consider *i* as an index for balancing service providers):

$$\underset{p_i^{\downarrow},\ p_i^{\uparrow}}{\text{Minimize}} \quad \sum_i \left(C_i^{\uparrow} p_i^{\uparrow} - C_i^{\downarrow} p_i^{\downarrow} \right)$$

subject to:

$$\sum_i \left(p_i^{\uparrow} - p_i^{\downarrow} \right) = \Delta P \quad : \quad \lambda^{\text{Balancing}}$$

$$0 \leq p_i^{\uparrow} \leq \overline{P}_i^{\uparrow} \quad \forall i$$

$$0 \leq p_i^{\downarrow} \leq \overline{P}_i^{\downarrow} \quad \forall i$$

For the sake of general formulation, load curtailment is treated as upward regulation (albeit at a very high cost)!

The balancing settlement

Day-ahead forecast: 20 MW
Offer price: €0/MWh

Capacity: 50 MW
Offer price: €20/MWh

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

Day-ahead market-clearing outcomes:

- Production level of W1: **20 MW**
- Production level of G1: **50 MW**
- Production level of G2: **20 MW**
- Consumption level of D1: **90 MW**
- Day-ahead market-clearing price: **€30/MWh**

The balancing settlement

Day-ahead forecast: 20 MW
Offer price: €0/MWh

Capacity: 50 MW
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Day-ahead market-clearing outcomes:

- Production level of W1: **20 MW**
- Production level of G1: **50 MW**
- Production level of G2: **20 MW**
- Consumption level of D1: **90 MW**
- Day-ahead market-clearing price: **€30/MWh**

Let's consider a case where

- Generator G1 fails in real-time, resulting in a 50 MW deficit, and
- The actual production of W1 is 35 MW (i.e., an extra 15 MW compared to the day-ahead schedule).

Therefore, the regulation need in the balancing stage is **50 - 15 = +35 MW.**

The balancing settlement

W1

Day-ahead forecast: 20 MW
Offer price: €0/MWh

G1 POWER PLANT

Capacity: 50 MW
Offer price: €20/MWh

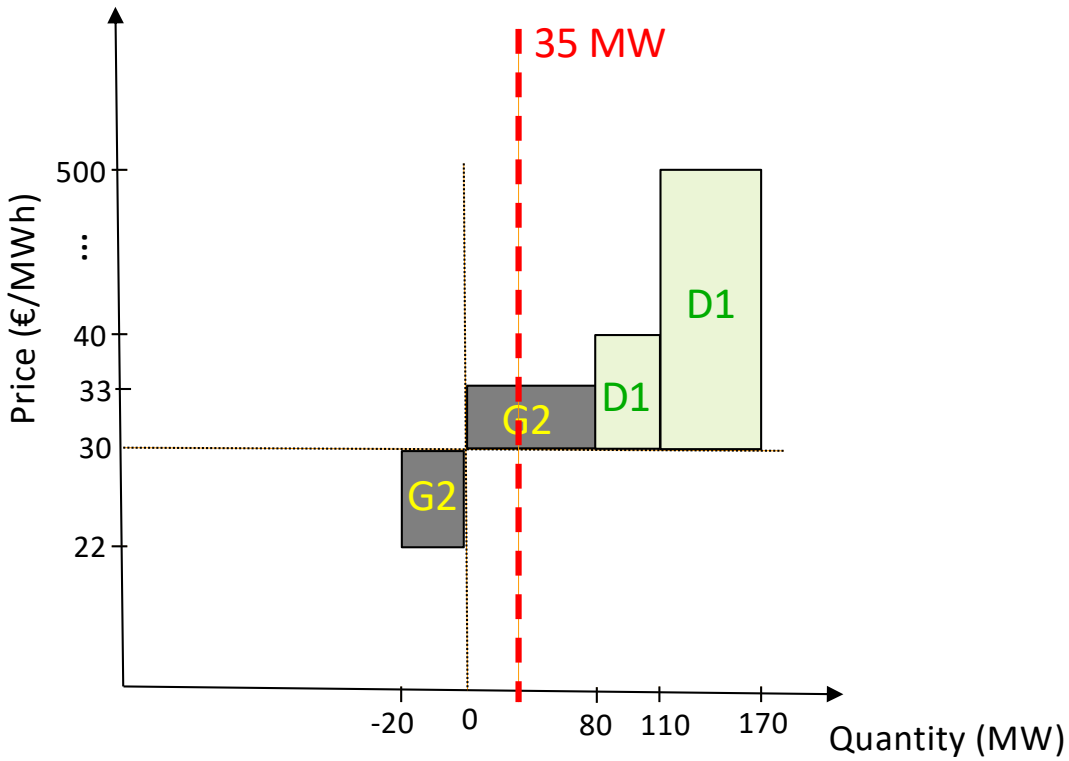
G2 POWER PLANT

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

The balancing need is 50-15 = +35 MW.



The balancing settlement

W1

Day-ahead forecast: 20 MW
Offer price: €0/MWh

G1 POWER PLANT

Capacity: 50 MW
Offer price: €20/MWh

G2 POWER PLANT

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

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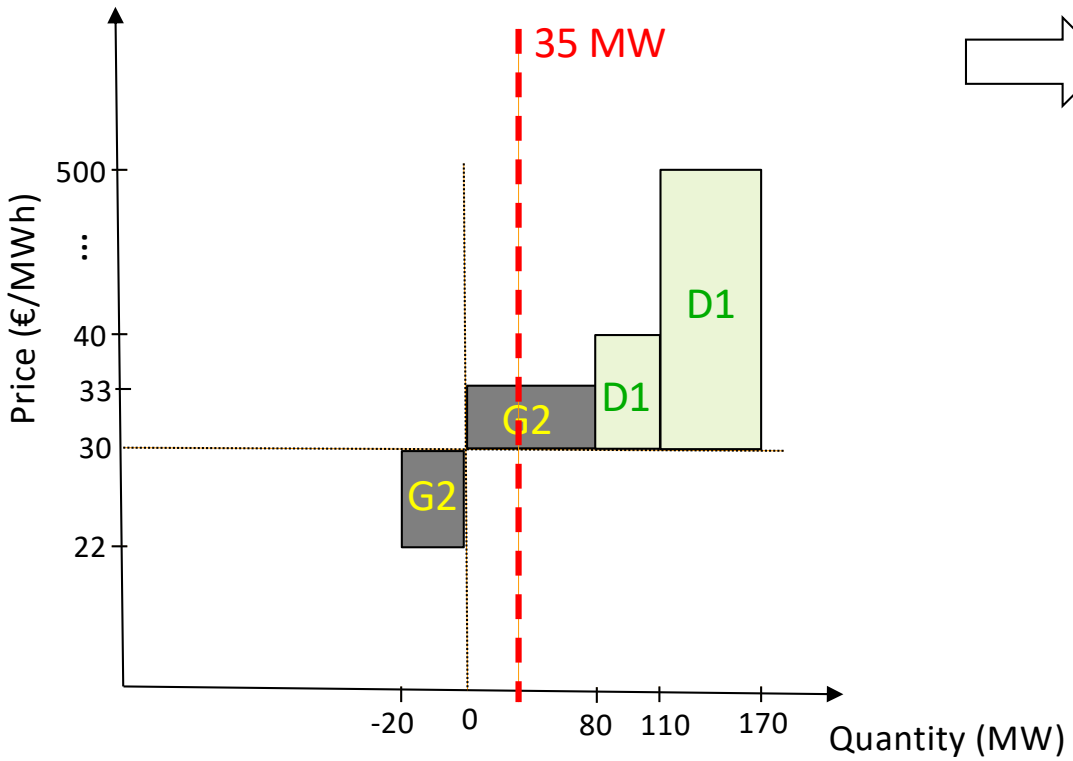


Balancing price = €33/MWh

The balancing settlement

W1

Day-ahead forecast: 20 MW
Offer price: €0/MWh

G1 POWER PLANT

Capacity: 50 MW
Offer price: €20/MWh

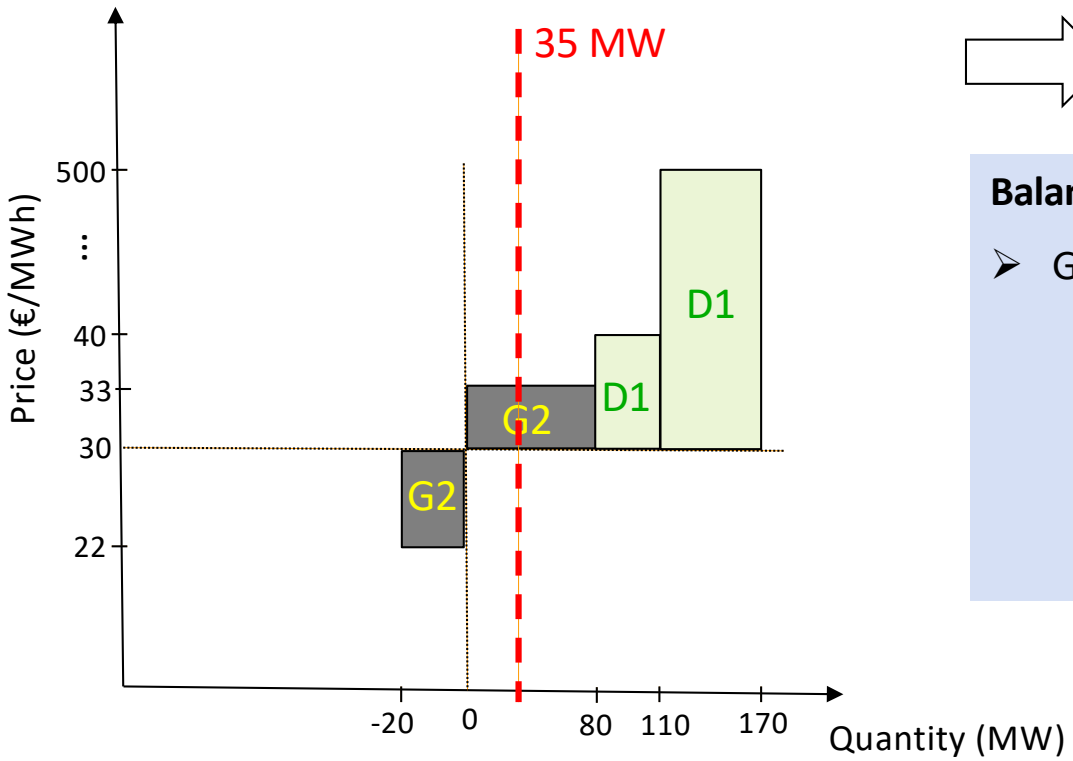
G2 POWER PLANT

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

The balancing need is 50-15 = +35 MW.



Balancing price = €33/MWh

Balancing service providers:

- G2 receiving 35 * 33 = €1,155

The balancing settlement

W1

Day-ahead forecast: 20 MW
Offer price: €0/MWh

G1 POWER PLANT

Capacity: 50 MW
Offer price: €20/MWh

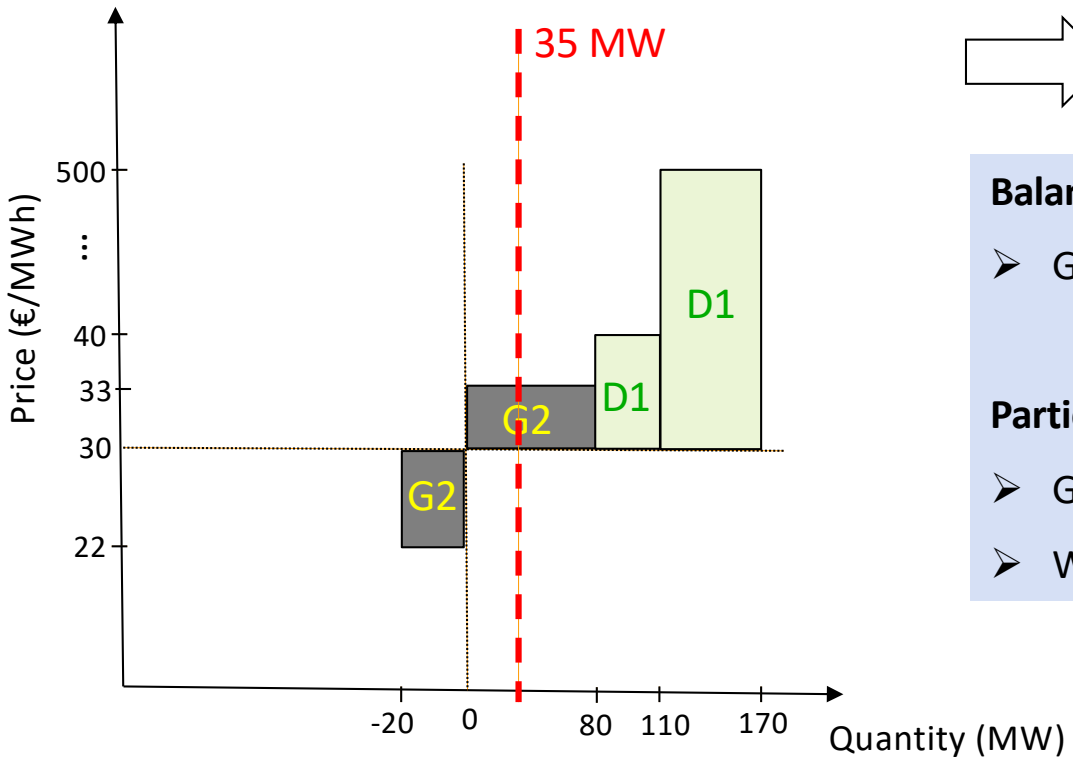
G2 POWER PLANT

Capacity: 100 MW
Offer price: €30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

The balancing need is 50-15 = +35 MW.



Balancing price = €33/MWh

Balancing service providers:

- G2 receiving 35 * 33 = €1,155

Participants who cause imbalance:

- G1 paying 50 * 33 = €1,650
- W1 receiving 15 * 33 = €495

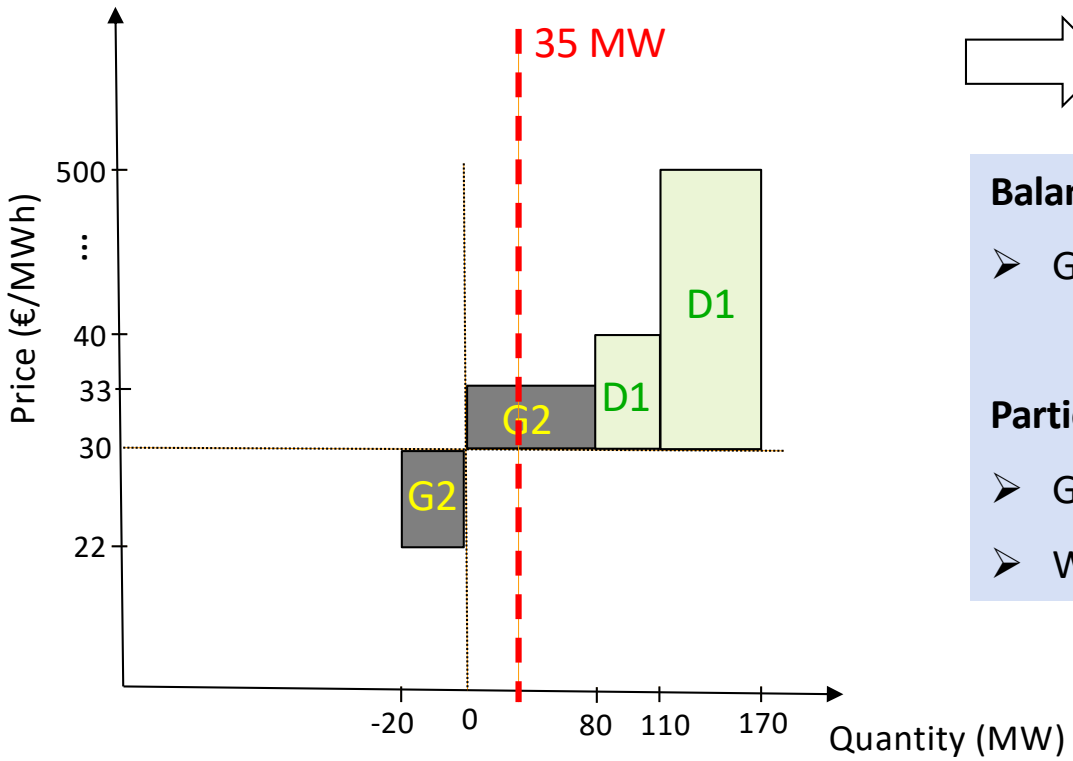
Total **revenues** in the day-ahead and balancing markets:

- G2 : [20 * 30] + [35 * 33] = €1,755
- G1 : [50 * 30] - [50 * 33] = - €150 (loss)
- W1 : [20 * 30] + [15 * 33] = €1,095
- Demand D1 pays [90 * 30] in the day-ahead market.

) MW
30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh



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- W1 : [20 * 30] + [15 * 33] = €1,095
- Demand D1 pays [90 * 30] in the day-ahead market.

MW
30/MWh

Demand D1

Maximum load: 90 MW
Bid price: €40/MWh

Two Remarks About Balancing Payments:

- The balancing settlement is **"budget balanced"** as G1 pays €1,650 while G2 and W1 together receive €1,650.
- W1 is paid at the same price as G2 even though W1 was not an intentional balancing service provider. Fortunately, its imbalance **"unintentionally"** helps the system by providing a **positive (desired) imbalance** when needed. However, this raises a key question: **Should W1 be paid at the balancing price of €33/MWh?**

Balancing price = €33/MWh

Balancing service providers:

- G2 receiving 35 * 33 = €1,155

Participants who cause imbalance:

- G1 paying 50 * 33 = €1,650
- W1 receiving 15 * 33 = €495

One-price versus two-price balancing schemes

- One-price is also referred to as "*single-price*"
- Two-price is also referred to as "*dual-price*"

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

In the **balancing** stage, we can consider four potential realizations:

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

In the **balancing** stage, we can consider four potential realizations:

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

The wind farm will be ***paid*** for its extra production in the balancing stage.

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

In the **balancing** stage, we can consider four potential realizations:

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

The wind farm will be ***charged*** for its shortage in production in the balancing stage.

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

In the **balancing** stage, we can consider four potential realizations:

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

For each realization, what do you expect for the balancing market price—higher or lower than the day-ahead price of €30/MWh?

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

In the **balancing** stage, we can consider four potential realizations:

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Under one-price scheme:

Higher than €30/MWh, because the system has a deficit and requires upward regulation!

Lower than €30/MWh, because the system has an excess and requires downward regulation!

Example

Assume:

- The day-ahead price is **€30/MWh**,
- The day-ahead schedule of a wind farm is **25 MW**.

In the **balancing** stage, we can consider four potential realizations:

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Under one-price scheme:

Example for a potential balancing price:

€33/MWh	€33/MWh	€27/MWh	€27/MWh

Example

Here, the wind farm will be paid a price **higher than** the day-ahead price of €30/MWh!

- Is there a potential motivation for the wind farm to *intentionally* withhold and not offer its most accurate forecast in the day-ahead market (assuming the farm correctly predicts that the system will be in a deficit condition)?

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW
Under one-price scheme:				
Example for a potential balancing price:	€33/MWh	€33/MWh	€27/MWh	€27/MWh

Example

Here, the wind farm will be paid a price **higher than** the day-ahead price of €30/MWh!

- Is there a potential motivation for the wind farm to *intentionally* withhold and not offer its most accurate forecast in the day-ahead market (assuming the farm correctly predicts that the system will be in a deficit condition)? Yes. How to remove this incentive?

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW
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Example for a potential balancing price:	€33/MWh	€33/MWh	€27/MWh	€27/MWh

Example

Here, the wind farm will be paid a price **higher than** the day-ahead price of €30/MWh!

- Is there a potential motivation for the wind farm to *intentionally* withhold and not offer its most accurate forecast in the day-ahead market (assuming the farm correctly predicts that the system will be in a deficit condition)? Yes. How to remove this incentive?

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Under one-price scheme:

Example for a potential balancing price: ~~€33/MWh~~ €33/MW

Although the market mechanism suggests a balancing price of €33/MWh, it is reduced to €30/MWh (the same as the day-ahead price).

- This is an **intervention** in the market-clearing mechanism!

Example

What about here?

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Under one-price scheme:

Example for a potential balancing price: €33/MWh €33/MWh €27/MWh €27/MWh

Example

Similarly, the wind farm may have a motivation to *intentionally* over-bid in the day-ahead market (being paid at €30/MWh in the day-ahead market and then paying back at €27/MWh in the balancing stage) as long as the farm correctly forecasts that the system will be in an excess condition!

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW
Under one-price scheme:				
Example for a potential balancing price:	€33/MWh	€33/MWh	€27/MWh	€27/MWh

Example

Similarly, the wind farm may have a motivation to *intentionally* over-bid in the day-ahead market (being paid at €30/MWh in the day-ahead market and then paying back at €27/MWh in the balancing stage) as long as the farm correctly forecasts that the system will be in an excess condition!

The power system has a	power deficit		power excess	
The realization of wind farm production i	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Although the market mechanism suggests a balancing price of €27/MWh, it is increased to **€30/MWh** (the same as the day-ahead price).

- This is again an **intervention** in the market-clearing mechanism!

Under one-price s

Example for a pote
balancing price:

MWh

~~€27/MWh~~

Example

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Under two-price scheme:

~~€33/MWh~~ €30/MWh	€33/MWh	€27/MWh	~~€27/MWh~~ €30/MWh

Example

Under the **two-price scheme**, there are two prices, one for each system condition (deficit or excess), with one of them being the same as the day-ahead price (here €30/MWh).

- Recall that this is an **intervention** in the market mechanism, which is not desirable.
- Instead, we should aim to increase the liquidity of the balancing market, meaning enhancing competitiveness by involving more market players.

The power system has a	power deficit		power excess	
The realization of wind farm production is	> 25 MW	< 25 MW	> 25 MW	< 25 MW

Under two-price scheme:

~~€33/MWh~~ €30/MWh	€33/MWh	€27/MWh	~~€27/MWh~~ €30/MWh

Takeaway for producers causing imbalance

The power system overall has a	Power deficit		Power excess	
Imbalance caused by a certain producer	Generation deficit	Generation excess	Generation deficit	Generation excess

Takeaway for producers causing imbalance

The power system overall has a	Power deficit		Power excess	
Imbalance caused by a certain producer	Generation deficit	Generation excess	Generation deficit	Generation excess
The imbalance caused by the producer is	Undesired	Desired	Desired	Undesired

- **Desired**: Although it is an unplanned imbalance, it helps the system by contributing to balancing the grid.
- **Undesired**: It is an unplanned imbalance that does not help the system and instead adds to the overall system imbalance.

Takeaway for producers causing imbalance

The power system overall has a	Power deficit		Power excess	
Imbalance caused by a certain producer	Generation deficit	Generation excess	Generation deficit	Generation excess

The imbalance caused by the producer is	Undesired	Desired	Desired	Undesired
One-price scheme	**Charged** at a price > DA price	**Paid** at a price > DA price	**Charged** at a price < DA price	**Paid** at a price < DA price
Two-price scheme	**Charged** at a price > DA price	**Paid** at a price = DA price	**Charged** at a price = DA price	**Paid** at a price < DA price

DA = day-ahead

Takeaway for demands causing imbalance

The power system overall has a	Power deficit		Power excess	
Imbalance caused by a certain demand	Extra consumption	Less consumption	Extra consumption	Less consumption

Takeaway for demands causing imbalance

The power system overall has a	Power deficit		Power excess	
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Takeaway for demands causing imbalance

The power system overall has a	Power deficit		Power excess	
Imbalance caused by a certain demand	Extra consumption	Less consumption	Extra consumption	Less consumption

The imbalance caused by the demand is	Undesired	Desired	Desired	Undesired
One-price scheme	**Charged** at a price > DA price	**Paid** at a price > DA price	**Charged** at a price < DA price	**Paid** at a price < DA price
Two-price scheme	**Charged** at a price > DA price	**Paid** at a price = DA price	**Charged** at a price = DA price	**Paid** at a price < DA price

DA = day-ahead

In Denmark and many other European countries

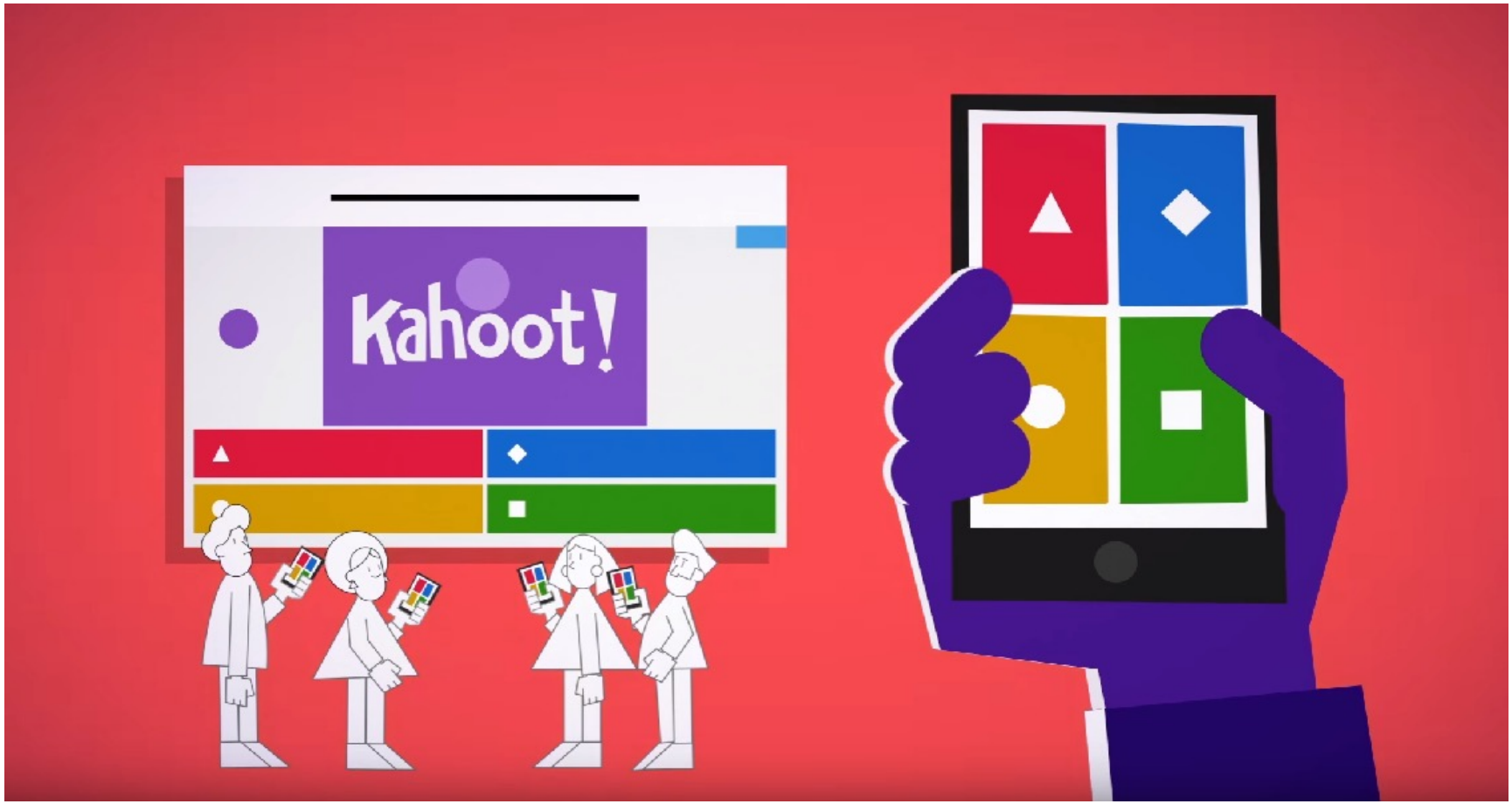
- The balancing mechanism is now based on a one-price scheme (previously it was a two-price scheme).

- Energinet switched to the one-price scheme in November 2021.

Where to find historical balancing data?

- Real-time imbalance data with a 1-minute resolution: https://www.energidataservice.dk/tso-electricity/PowerSystemRightNow

- Balancing prices: https://www.energidataservice.dk/tso-electricity/RegulatingBalancePowerdata


Kahoot!

Thanks for your attention!

Email: jalal@dtu.dk